# Dilan Karaguler

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## Research Interests

Applied algebraic topology and topological data analysis; path homology and Mayer (N-nilpotent) homology of directed graphs and path complexes; persistent homology; topology-informed graph machine learning and geometric deep learning, with applications to molecular, dynamical and engineering data.

## Education

**Michigan State University** — East Lansing, MI, US
Ph.D. in Mathematics (Candidate) — August 2022 – May 2027 (Expected)
- Advisors: Prof. Ekaterina Rapinchuk and Prof. Guowei Wei

**Middle East Technical University (METU)** — Ankara, Türkiye
M.S. in Mathematics — September 2019 – September 2021
- Thesis: *A Survey on Multidimensional Persistence Theory*
- Advisor: Assoc. Prof. Semra Pamuk

**Middle East Technical University (METU)** — Ankara, Türkiye
B.S. in Mathematics, Honor Student — June 2014 – June 2019

## Publications

**D. Karaguler** and G.-W. Wei. *Mayer Path Homology.* Foundations of Data Science, accepted (2026). doi:10.3934/fods.2026025, arXiv:2605.16525.

**D. Karaguler.** *A Survey on Multidimensional Persistence Theory.* M.S. Thesis, Middle East Technical University (2021).

**D. Karaguler** and G.-W. Wei. *Persistent Mayer Path Laplacians.* In preparation (2026).

## Research Experience

**Graduate Researcher** — August 2022 – Present
Michigan State University — East Lansing, MI, US
- Develop Mayer path homology, a homology theory for directed graphs built from N-nilpotent differentials on path complexes; published in *Foundations of Data Science* (with G.-W. Wei).
- Implement topological feature extraction (path complexes, Mayer homology and Laplacians, persistent homology) in Python with GUDHI, NumPy and PyTorch, and combine these features with graph neural networks.

**Research Mentor** — May 2023 – June 2023
Topology RTG, Michigan State University — East Lansing, MI, US
- Mentored an undergraduate researcher on a project in Morse theory: project design, methodology and research presentation; the student went on to major in mathematics.

## Industry Experience

**Software Engineer Intern** — May 2026 – August 2026
Intel Corporation — Oregon, US
- Built an end-to-end graph transformation pipeline in Python that extracts schematic topology directly from chip layout files, without pre-existing schematic metadata.
- Modeled low-level device instances as graph nodes and extracted a local subgraph around each instance to capture its connectivity.
- Developed graph aggregation and embedding routines producing a vector representation per device, used to classify components as analog, digital or fill devices.
- Worked with Design Yield and Manufacturability engineers to evaluate the prototype for automated structural recognition in layout verification.

**Graduate Technical Intern** — May 2025 – August 2025
Intel Corporation — Oregon, US
- Performed spatial pattern analysis on semiconductor chip layer images with KLayout and Python to identify and cluster recurring structures across layout datasets.
- Applied graph neural networks to search for and classify similar patterns across multi-layer designs.
- Partnered with cross-functional yield teams to bring ML-based pattern recognition into automated yield-analysis workflows.

## Teaching Leadership & Mentoring

**Lead Teaching Assistant** August 2024 – Present
Department of Mathematics, Michigan State University East Lansing, MI, US

- **2026 – 2027:** Co-lead, with Eloy Moreno-Nadales, a yearlong workshop series that prepares first-year graduate students for their first teaching assignments. Sessions so far: an introduction run as a gallery walk, with a "Math PhD passport" and department scavenger hunt connecting new students with senior graduate students; and *Three Aspects of Teaching* (clarity and delivery, communication and language, engagement and cohort needs), built on the TAs' own classroom observations to prepare their mock teaching.
- **Fall 2025 (MTH 299):** Prepared recitation worksheet materials, gathered TA feedback for the instructors, and substituted for recitation leaders when needed.
- **Spring 2025 (MTH 133):** Ran TA meetings, prepared weekly recitation worksheets, and set up Teams tools to track the TA timeline and grading deadlines.
- **Fall 2024:** Ran the orientation for undergraduate learning assistants (ULAs) starting in August, held mock-teaching and feedback sessions, and observed ULAs and GTAs during the semester to give feedback; as MTH 133 pod leader, ran meetings with recitation leaders and prepared weekly recitation worksheets.

**Graduate Lead, Center for Instructional Mentoring (CIM)** Summer 2024
Department of Mathematics, Michigan State University East Lansing, MI, US

- Mentored first-time graduate instructors through weekly check-ins, one-on-one conversations and non-evaluative teaching feedback, and introduced the teaching technology they could use.
- Created a teaching handbook built on interviews with senior GTAs about the courses they taught, describing each course's expectations so junior graduate students can choose assignments that fit their needs.

**Mentor, Academic Culture and English Skills (ACES) Program** Summer 2024
Michigan State University East Lansing, MI, US

- Mentored incoming international graduate students preparing to become MSU teaching assistants, introducing U.S. classroom culture, university structure and cross-cultural communication, and organized social events.

**Supervisor, Math Learning Center (MLC)** Fall 2022 – Spring 2023, Fall 2024
Michigan State University (Teaching Exemption Fellowship, 2022 – 2023) East Lansing, MI, US

- Managed how tutors were distributed among students and arranged the room to match demand.

**Student feedback:** dilankaraguler.github.io/teaching/what-students-say

## Courses Taught

**MTH 314 — Matrix Algebra with Computational Applications** Fall 2026
Instructor (1 section), Michigan State University East Lansing, MI, US

- Prepare and deliver lectures, facilitate group work, and hold help-room hours.

**MTH 299 — Transitions** Fall 2025
Teaching Assistant (2 sections), Michigan State University East Lansing, MI, US

- Co-taught with Ester Sgallova: led weekly recitations, wrote homework solutions, graded homework and quizzes, and held help-room hours.

**MTH 314 — Matrix Algebra with Computational Applications** Spring 2025
Teaching Assistant (1 section), Michigan State University East Lansing, MI, US

- Graded homework and the final exam, ran quiz and exam review sessions, and held help-room hours.

**MTH 133 — Calculus II** Fall 2024
Teaching Assistant (1 section) and pod leader, Michigan State University East Lansing, MI, US

- Led a recitation section; as pod leader, ran meetings with the recitation leaders and prepared the weekly recitation worksheets.

**MTH 235 — Differential Equations** Spring 2024
Teaching Assistant (3 sections), Michigan State University East Lansing, MI, US

- Wrote recitation notes with lecture summaries and practice problems, graded quizzes, held help-room hours, and followed up with students in online sections.

**MTH 133 — Calculus II** Fall 2023
Teaching Assistant (3 sections: 2 online, 1 in person), Michigan State University East Lansing, MI, US

- Prepared weekly lecture summaries for the recall session before group work and worksheets; graded and proctored worksheets and quizzes.
- Held help-room hours and tracked student progress, reaching out to students as needed, especially in online sections.

**MTH 133 — Calculus II** Summer 2023
Teaching Assistant (1 online asynchronous section), Michigan State University East Lansing, MI, US

– Contacted students weekly by email and tracked their course activity.

**MTH 154, 153, 120, 119 — Calculus I and II** January 2020 – June 2022
Graduate Teaching Assistant, Middle East Technical University Ankara, Türkiye

– MTH 154 Calculus II for Mathematics Students (Spring 2022), MTH 153 Calculus I for Mathematics Students (Fall 2021), MTH 120 Calculus II for Engineering Students (Spring 2021), MTH 119 Calculus I for Engineering Students (Fall 2020).
– Led recitations and office hours, graded, and proctored online and in-person exams.

**Calculus for Mathematics Students** March 2018 – June 2019
Undergraduate Teaching Assistant, Middle East Technical University Ankara, Türkiye

– Graded homework and held review hours.

## Teaching Development & Educational Research

**Program Evaluator & Instructional Researcher** May 2026
AI for Advanced Semiconductor Manufacturing Workshop, Michigan State University East Lansing, MI, US

– Designed pre/post-workshop surveys and observation protocols to evaluate program effectiveness and adult learning outcomes.
– Analyzed participant feedback quantitatively and with qualitative thematic coding, tracking confidence growth across AI and semiconductor manufacturing topics.
– Wrote an evaluation report grounded in adult learning theory with curriculum recommendations for the workshop.

**Colleges Online Learning Academy (COLA) Scholar** Summer 2024
Michigan State University Portfolio

– Built a teaching portfolio website and wrote a project plan for online course design as program deliverables.

## Talks

**Integrating AI to Lecture Prep (upcoming)** November 2026
Mathematics Teaching Seminar, *Sparks: Teaching Lightning Talks by Graduate Instructors* MSU

**Mayer Path Homology** April 2026
TDA Seminar, Dept. of Computational Mathematics, Science and Engineering, MSU East Lansing, MI, US

**Interaction Topology in TDA** October 2025
Invited lecture, MTH 994: Mathematics-Assisted AI and AI-inspired Mathematics, MSU East Lansing, MI, US

**Mayer Topology in TDA** October 2025
Invited lecture, MTH 994: Mathematics-Assisted AI and AI-inspired Mathematics, MSU East Lansing, MI, US

**From Data to Barcodes** April 2021
Graduate Student Seminar, Middle East Technical University Ankara, Türkiye

## Poster Presentations

***Mayer Path Topology and Persistence*** September 2025
SIAM Great Lakes Section Meeting, Illinois Institute of Technology Chicago, IL, US

***Mayer Path Topology and Persistence*** March 2025
Mathematics of Data, Dynamics, and Life Sciences Conference, UC Irvine Irvine, CA, US

## Projects

**Persistent Mayer Homology with Dynamic Mode Decomposition** April 2026
Python, NumPy, SciPy, GUDHI GitHub

– Built a reproducible pipeline combining Vietoris–Rips filtrations, Mayer homology and Dynamic Mode Decomposition (DMD) to track time-varying topological features.
– Computed Mayer Betti trajectories on synthetic 3D point dynamics; a rank-10 DMD model reconstructed a 500-frame dataset with about 3.6% relative error and supported multi-step forecasting.

**Homology-Fed Neural Network Framework** November 2024
Python, PyTorch, GUDHI, NumPy GitHub

– Wrote a Python library computing Mayer homology and path-complex Laplacians on graph-structured data.
– Used the resulting topological features for molecular property prediction on the QM9 dataset with graph neural networks.

**Cellular Automata on Curved Surfaces** March 2024
Python, NumPy, Matplotlib GitHub

– Simulated Conway's Game of Life on a square, torus and sphere with surface-specific neighbor rules, and analyzed how board size and initial density affect population persistence.

## Awards, Fellowships & Scholarships

Doctoral Student Summer Internship Fellowship, The Graduate School, Michigan State University — Summer 2026

**Excellence in Teaching Award**, Department of Mathematics, Michigan State University — May 2025

Colleges Online Learning Academy (COLA) Fellowship, Michigan State University — Summer 2024

Teaching Exemption Fellowship, Michigan State University — 2022 – 2023

RONESANS Scholarship, RONESANS — 2018 – 2019

UNIGENC Scholarship, UNIGENC Association — 2015 – 2019

Merit Scholarship, Turkish Ministry of National Education — 2010 – 2014

## Service, Leadership & Outreach

**Graduate Student Representative, Graduate Studies Committee** — 2025 – 2026
Department of Mathematics, Michigan State University — East Lansing, MI, US

- Represented all mathematics graduate students at committee meetings with faculty.
- Planned and prepared Responsible Conduct of Research (RCR) workshops with the Graduate Chair.
- Met with prospective graduate students and hosted graduate student town hall meetings on the state of the department.
- Served as a point of contact for graduate students with grievances and directed them to department and university resources.

**President / Board Member** — August 2022 – August 2025
Association for Women in Mathematics (AWM), MSU Chapter — East Lansing, MI, US

- Organized seminars, career workshops, informational sessions on academic life and social events for women in mathematics.

**Member, First-Generation/Low-Income (FGLI) Group** — September 2024 – May 2025
Department of Mathematics, Michigan State University — East Lansing, MI, US

- Took part in community gatherings and social events with first-generation and low-income undergraduates.

**Organizing Committee Member** — April 2024
Graduate Student Topology and Geometry Conference, Michigan State University — East Lansing, MI, US

- Handled communication with speakers and attendees, room assignments and talk scheduling, and moderated sessions.

**President** — February 2015 – December 2018
METU Math Club — Ankara, Türkiye

- Organized seminars, workshops and regular reading groups in topology, algebra and other areas.
- Led Math Club reading sessions with first-year students in abstract algebra and number theory, preparing each student to lead a discussion of part of the material.

## Conferences & Workshops Attended

Agentic AI (hot topic course), AIDMM NSF Research Traineeship, Michigan State University — Fall 2026

AI, Data and PDE Workshop, Michigan State University — April 2025

Graduate Student Topology and Geometry Conference, Michigan State University — April 2024

Mid-Atlantic Topology Conference, Northeastern University — March 2024

SIAM Great Lakes Section Meeting, Michigan State University — October 2023

Computational Persistence, Purdue University — September 2023

Randomness in Topology and its Applications, IMSI, Chicago — March 2023

LMS Workshop: Applied Algebraic Topology, Queen Mary University of London (online) — February 2022

Topological Insights in Neuroscience, IMSI (online) — May 2021

Topological Data Analysis, IMSI (online) — April 2021

## Technical Skills

**Programming**: Python, R, SQL
**Methods**: Graph Neural Networks, Graph Transformers, Topological Data Analysis, Persistent Homology, Vietoris–Rips Filtrations, Dynamic Mode Decomposition, Supervised/Unsupervised Learning, Model Evaluation
**Libraries**: PyTorch, PyTorch Geometric, NetworkX, GUDHI, NumPy, SciPy, Pandas, Scikit-Learn, TensorFlow, Keras, Matplotlib, XGBoost, LightGBM
**Tools**: Git, Linux/Unix, DuckDB, SQLite, Jupyter, KLayout, LaTeX
**Domain Knowledge**: Design for Manufacturability (DFM), Semiconductor Yield Analysis, Schematic Extraction, Chip Layout Analysis

## References

Available upon request.